Department of Computer Engineering

HiTec University, Taxila

Complex Engineering Problem

Digital Image Processing (CS-406)

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Complex Engineering Problem
Automated License Plate Recognition (ALPR)

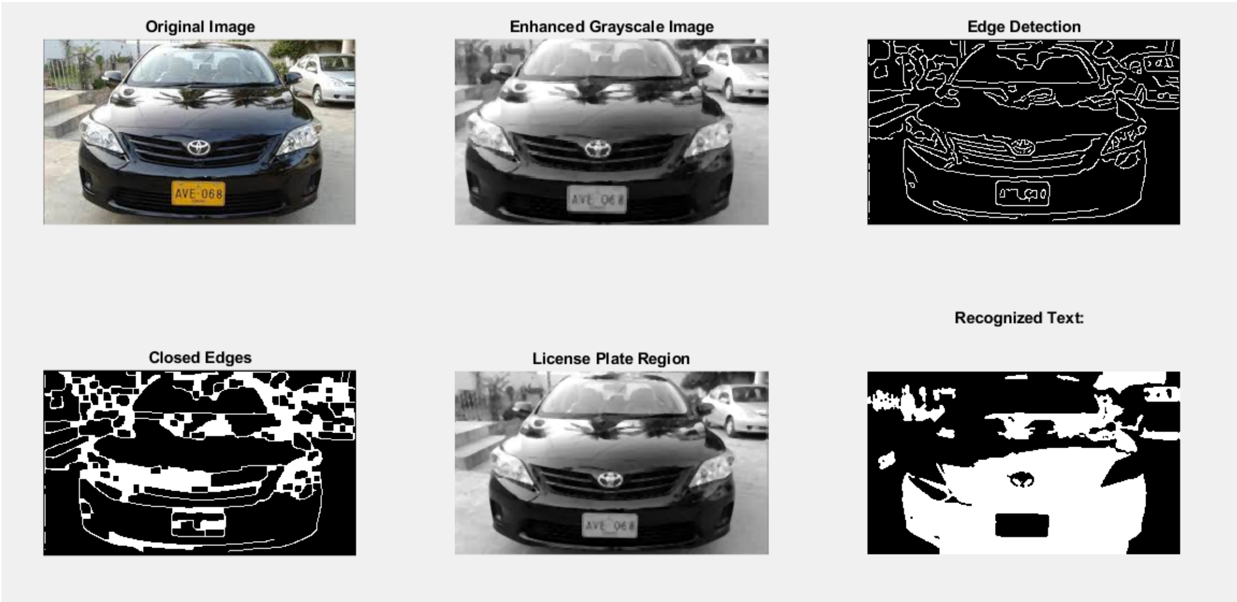
Develop a MATLAB program for Automated License Plate Recognition (ALPR) to assist in traffic law enforcement. Your program should process an input image of a vehicle, identify the license plate, extract its characters, and display them as text.

MATLAB CODE:

```
% Load the image
imagePath = 'CAR_IMAGE.jpg'; % Path to the uploaded image
originalImage = imread(imagePath);
% Pre-processing
grayImage = rgb2gray(originalImage); % Convert to grayscale
enhancedImage = imadjust(grayImage); % Enhance contrast
% Edge Detection
edges = edge(enhancedImage, 'Canny'); % Detect edges
% Region of Interest Interaction (License Plate Extraction)
regions = regionprops(edges, 'BoundingBox', 'Area');
[~, largestRegionIndex] = max([regions.Area]); % Find largest region
boundingBox = regions(largestRegionIndex).BoundingBox;
% Extract the license plate
licensePlate = imcrop(enhancedImage, boundingBox);
% Character Segmentation
binarizedPlate = imbinarize(licensePlate); % Convert to binary
binarizedPlate = ~bwareaopen(~binarizedPlate, 30); % Remove small
objects
characterRegions = regionprops(binarizedPlate, 'BoundingBox');
% Extract individual characters
```

```
for i = 1:length(characterRegions)
    charBox = characterRegions(i).BoundingBox;
    characterImage = imcrop(binarizedPlate, charBox);
    % Resize for OCR processing
    resizedChar = imresize(characterImage, [42, 24]);
    segmentedCharacters{i} = resizedChar; %#ok<SAGROW>
end
% Optical Character Recognition (OCR)
recognizedText = ocr(licensePlate, 'CharacterSet',
'ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789');
disp(['Recognized License Plate: ', recognizedText.Text]);
% Display Results
figure;
subplot(2, 3, 1), imshow(originalImage), title('Original Image');
subplot(2, 3, 2), imshow(enhancedImage), title('Enhanced Grayscale
Image');
subplot(2, 3, 3), imshow(edges), title('Edge Detection');
subplot(2, 3, 4), imshow(licensePlate), title('License Plate Extracted');
subplot(2, 3, 5), imshow(binarizedPlate), title('Binarized License Plate');
subplot(2, 3, 6), imshow(licensePlate), title(['Recognized Text: ',
recognizedText.Text]);
```

MATLAB OUTPUT:


Original Image
Enhanced Grayscale Image
Edge Detection
Closed Edges
License Plate Region
Recognized Text: